

A

Explanation Templates

Our explanation generation system uses predefined templates organized by trigger type. Each trigger category contains multiple template variations to provide natural language diversity and avoid repetitive messaging. Templates use placeholder variables that are dynamically filled based on the agent's current subtask and environmental context.

A.1

Primary Explanation Templates

The following templates were used in the main study (conducted in English). Each template includes placeholder variables that are replaced with context-specific information during explanation generation.

A.1.1

Blocking Trigger Templates

Used when the human participant blocks the agent's intended movement path:

- “Moving around you to {task}”
- “Excuse me, I want to {task}”

A.1.2

Distance Threshold Trigger Templates

Used when the agent moves a significant distance (layout-dependent threshold) since the last explanation:

- “Moving to {task}”
- “Going to {task}”
- “Heading to {task}”

A.1.3

Critical Path Trigger Templates

Used when the agent performs a subtask crucial to goal completion that would be counterproductive for the teammate to duplicate:

- “I need {object}”
- “Getting {object}”
- “I’ll grab {object}”

A.1.4

Subtask Change Trigger Templates

Used when the agent transitions to a new subtask (excluding transitions to “unknown” subtask):

- “I want to {task}”
- “Now I’ll {task}”

A.2

Fallback Templates

When the agent’s current subtask is unknown or undefined, simplified blocking explanations are used that do not reference specific tasks:

- “Excuse me, I need to get past”
- “Trying to get around you”
- “You’re blocking my path”

A.3

Template Variable Mappings

A.3.1

Subtask Descriptions ({task} placeholder)

The following readable descriptions are used to replace the {task} placeholder based on the agent's current subtask:

- get_onion_from_dispenser → "grab an onion from dispenser"
- get_onion_from_counter → "grab an onion from the counter"
- put_onion_in_pot → "put onion in pot"
- put_onion_closer → "place onion on the counter"
- get_plate_from_dish_rack → "grab dish from dispenser"
- get_plate_from_counter → "grab dish from the counter"
- put_plate_closer → "place dish on the counter"
- get_soup → "get the soup"
- get_soup_from_counter → "grab soup from counter"
- put_soup_closer → "place soup on the counter"
- serve_soup → "serve the soup"
- wait → "wait"

A.3.2

Destination Descriptions (for Distance Trigger)

When using distance threshold triggers, destinations are described based on subtask context:

- get_onion_from_dispenser → "the onion dispenser"
- get_onion_from_counter → "get an onion from the counter"
- put_onion_in_pot → "the cooking pot"
- put_onion_closer → "place an onion near the pot"
- get_plate_from_dish_rack → "the dish rack"
- get_plate_from_counter → "get a plate from the counter"
- put_plate_closer → "place a plate near the pot"
- get_soup → "get soup from the pot"
- get_soup_from_counter → "get soup from the counter"
- put_soup_closer → "place soup closer to serving area"
- serve_soup → "the serving area"

A.3.3

Critical Path Objects ({object} placeholder)

For critical path triggers, specific objects are referenced based on game state:

- “the final onion” (when a pot needs only one more onion to start cooking)
- “a plate” (when soup is ready but needs serving)
- “this soup” (when delivering completed soup)

A.4

Implementation Notes

- Templates are randomly selected from available options within each trigger category to provide variation;
- All explanations are subject to a 6-second cooldown period to prevent information overload;
- In the Text condition, each explanation remains visible for 4 seconds;
- Distance thresholds for triggering explanations vary by layout: cramped_room (4 tiles), coordination_ring (5 tiles), counter_circuit (6 tiles), with a default of 5 tiles for other layouts – though only the Counter Circuit layout was used for this study.

B

Surveys

The complete question lists of all survey instruments used in our study are available below.

B.1

Demographics Survey

1. What is your age? (*Open numeric response*)
2. What gender do you identify as?
 - Male
 - Female
 - Non-binary
 - I prefer not to answer
3. What is your highest level of education?
 - Some elementary/primary school
 - Completed elementary/primary school
 - Some high school
 - Completed high school
 - High school graduate
 - Some college/university
 - Bachelor's degree
 - Graduate degree
4. What's your field of work?
 - Engineering
 - Computer Science/IT
 - Natural Sciences (Mathematics, Physics, etc)
 - Humanities
 - Biological Sciences/Health
 - Arts/Design
 - Other
5. Do you have any academic or professional experience with robotics or artificial intelligence?
 - Yes, professional
 - Yes, academic
 - Yes, as a hobby
 - No

B.2**Videogame Experience Survey**

1. On a scale from 0 to 10, how would you rate your familiarity with electronic games (videogames)? (*0 = Not familiar at all, 10 = Very familiar*)
2. On average, how many hours per week do you play videogames?
 - I don't play videogames
 - 1-3 hours
 - 4-7 hours
 - 8-14 hours
 - 15+ hours
3. Have you ever played videogames that involve cooperation with computer-controlled characters (NPCs/AI)?
 - Yes, frequently
 - Yes, occasionally
 - Yes, rarely
 - Never
4. Have you played Overcooked or Overcooked 2 before?
 - Yes, I've played extensively (more than 20 hours)
 - Yes, I've played (5-20 hours)
 - Yes, a few times (less than 5 hours)
 - Never
5. Are you familiar with Overcooked-AI? (Descriptive text: *A version of Overcooked adapted for research; knowing it is not a requirement.*)
 - Yes, I've participated in studies that use it
 - Yes, I've played it
 - Yes, but I've never played it
 - No

B.3**Nasa-TLX Workload Assessment**

All questions rated on a scale from 1 to 7. For questions 1-3, 5-6: 1 = "Very Low", 7 = "Very High". For question 4: 1 = "Perfect", 7 = "Failure".

1. **Mental Demand:** How mentally demanding was the task?
2. **Physical Demand:** How physically demanding was the task?
3. **Temporal Demand:** How hurried or rushed was the pace of the task?
4. **Performance:** How successful were you in accomplishing what you were asked to do?
5. **Effort:** How hard did you have to work to accomplish your level of performance?
6. **Frustration:** How insecure, discouraged, irritated, stressed, and annoyed were you?

B.4**Human-Agent Fluency Assessment**

All items (except the last question) rated on a 7-point scale from 1 (Strongly Disagree) to 7 (Strongly Agree).

1. The human-agent team worked fluently together.
2. The fluency of the human-agent team improved over time.
3. The agent contributed to the fluency of the interaction.
4. I had to carry most of the workload to make the human-agent team successful.
5. I had to take charge to ensure the human-agent team performed well.
6. The agent contributed equally to the team performance.
7. I was the most important team member.
8. The agent was the most important team member.
9. I trusted the agent would do the right task at the right time.
10. The agent was trustworthy.

11. The agent was intelligent.
12. The agent was committed to the task.
13. The human-agent team improved over time.
14. The fluency of the human-agent team improved over time.
15. The agent's performance improved over time.
16. I felt uncomfortable with the agent.
17. The agent and I understood each other.
18. I believe the agent likes me.
19. The agent and I respect each other.
20. I am confident in the agent's ability to help me.
21. I feel the agent appreciates me.
22. The agent and I trust each other.
23. The agent perceived correctly what my goals were.
24. The agent did not understand what I was trying to accomplish.
25. The agent and I are working towards mutually agreed upon goals.
26. The agent made important contributions to the team's success.
27. The agent was committed to the team's success.
28. I was committed to the team's success.
29. The agent was cooperative.
30. Feel free to add any comments related to the agent or your experience with it in the field below. (*Open text response*)

B.5**Godspeed Questionnaire**

Participants were given the following instructions: "Please mark the scales below according to your perceptions of the agent throughout the game sessions. If the agent seemed more like the characteristic on the left, mark a lower number. If it seemed more like the characteristic on the right, mark a higher number." All items were rated on a scale from 1 to 5, where the first word represented the leftmost label and the second represented the rightmost label.

B.5.1**Anthropomorphism**

1. Fake – Natural
2. Machine-like – Human-like
3. Unconscious – Conscious
4. Artificial – Lifelike
5. Moving rigidly – Moving elegantly

B.5.2**Animacy**

6. Dead – Alive
7. Stagnant – Lively
8. Mechanical – Organic
9. Artificial – Lifelike
10. Inert – Interactive
11. Apathetic – Responsive

B.5.3**Likeability**

12. Dislike – Like
13. Unfriendly – Friendly
14. Unkind – Kind
15. Unpleasant – Pleasant
16. Awful – Nice

B.5.4**Perceived Intelligence**

17. Incompetent – Competent
18. Ignorant – Knowledgeable
19. Irresponsible – Responsible
20. Unintelligent – Intelligent
21. Foolish – Sensible

B.5.5**Perceived Safety**

22. Anxious – Relaxed
23. Calm – Agitated
24. Quiescent – Surprised

B.6**Instructional Manipulation Check Questions**

The following questions were embedded within surveys to assess participant attention and data quality. Participants were expected to select the specific number mentioned in each question.

1. To demonstrate that you are reading the instructions, please select 6 for this question. (*Correct answer: 6*)
2. Please select 2 to show you are paying attention to the survey. (*Correct answer: 2*)
3. To verify you are following along, please select 4 in the slider below for this question. (*Correct answer: 4*)
4. Please select 1 in the slider below to confirm you are reading each question carefully. (*Correct answer: 1*)

IMC Question Distribution:

- Questions 1 and 2: Included in the Human-Agent Fluency Assessment and embedded within the 1st and 3rd NASA-TLX workload surveys.
- Questions 3 and 4: Included in the Godspeed Questionnaire.

Note: Participants achieving less than 80% accuracy across all IMC questions were excluded from the final analysis to ensure data quality.